

A Duration-Neutral Issuer-Curve Strategy

WRDS enhanced end-of-month bond data | August 2022 to March 2025

0.39% Gross annualized return	1.02% Annualized volatility	-1.98% Annualized return after 25 bp cost
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FINAL FINDING

The concentrated top-two portfolio produces the strongest result of the three specifications, but its expected return is too small to overcome turnover costs. In this sample, issuer-curve residuals are useful for ranking relative value; they do not provide evidence of implementable trading alpha.

Course	Instructor	Sample	Issuers
MTH 9897 - Systematic Trading	Professor Misha Boroditsky	Aug. 2022 to Mar. 2025	AAPL, AMZN, BA, CAT, DIS, T

Prepared from the executed assignment notebook and two supplied WRDS files.

Executive Summary

BOTTOM LINE
Pass 2 earns 0.39% gross annualized return with 1.02% volatility and a 0.38 gross Sharpe ratio. After a 25 bp one-way turnover charge, annualized return is -1.98% and Sharpe is -1.96.

I estimate issuer-specific spread curves for six large corporate issuers, rank bonds by actual spread minus fitted spread, and form monthly long-cheap/short-rich portfolios. Each issuer sleeve is scaled to approximately zero net DV01 and then combined with equal issuer weights.

Results across the three specifications

Portfolio	Gross ann. return	Gross vol.	Gross Sharpe	Avg. turnover	Net ann. return (25 bp)
1. Baseline top 3	0.15%	0.78%	0.19	0.78	-2.10%
2. Concentrated top 2	0.39%	1.02%	0.38	0.82	-1.98%
3. Liquidity-aware top 3	0.10%	0.64%	0.15	0.79	-2.17%

How I interpret the evidence

Concentrating on the two strongest residuals improves the gross result relative to the baseline. Adding liquidity to the ranking does not improve performance in the main sample. More importantly, all three designs become negative under the 25 bp cost assumption, so portfolio selection does not overcome implementation frictions.

1.00% Pass 2 gross total return	\$10,067 Sum of monthly gross PnL on \$1mm	31 Monthly observations
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The economic conclusion is narrow: the model organizes cross-sectional bond opportunities and produces a mechanically sound rate hedge, but the observed edge is not large enough to support a tradable strategy after plausible costs.

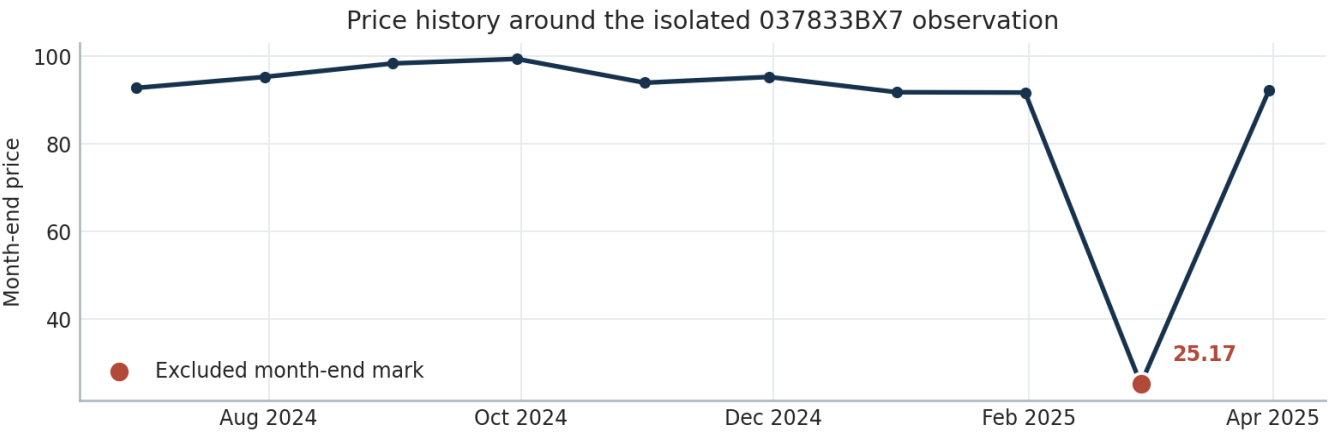
1. Data and Cleaning

The sample uses enhanced end-of-month observations for six large non-financial issuers. I require at least one year to maturity and approximately \$200 million outstanding, remove defaulted and convertible observations, and keep the modal security level and rating category within each issuer-month.

Issuer	Observations	Bonds	Median duration	Duration range	Median spread
AAPL	1,474	52	7.31	0.95 - 20.99	19.0 bps
AMZN	908	33	6.58	0.95 - 20.03	15.0 bps
BA	1,304	51	7.39	0.92 - 16.41	28.5 bps
CAT	957	51	3.45	0.95 - 18.65	13.0 bps
DIS	1,237	44	8.82	0.96 - 18.19	20.0 bps
T	1,678	56	10.63	0.96 - 17.84	28.0 bps

Price-quality rule

Before fitting curves or constructing returns, I flag an isolated month-end mark when it differs by more than 40% from both adjacent monthly prices, the neighboring prices are within 20% of one another, and both date gaps are between 25 and 45 days. The rule identifies one observation.



EXCLUDED OBSERVATION
AAPL CUSIP 037833BX7 is recorded at 91.76, 25.17, and 92.24 across January, February, and March 2025. I remove the February mark before both model fitting and return construction. The resulting 59-day January-to-March interval is not treated as a monthly return.

2. Return and Curve Model

Return construction

Positions are formed at month-end and evaluated with the next observation for the same bond when the date gap is 25 to 45 days. WRDS **RET_EOM** is the primary total-return measure. If it is unavailable, the fallback combines clean-price return with coupon accrual using actual gap days over 360, consistent with the daily PnL convention Professor Boroditsky discussed in class.

MONTHLY RETURN FALLBACK

Price return + annual coupon rate x gap days / 360

Issuer-curve specification

For each issuer-month, I fit the assignment's transparent three-factor spread model. The regression weights combine TRACE dollar volume and amount outstanding, with both inputs winsorized within issuer-month so a single large bond cannot dominate calibration.

CURVE EQUATION

T_Spread = b0 + b1 log(1 + Duration) + b2 Duration + b3 Coupon

Fit diagnostics

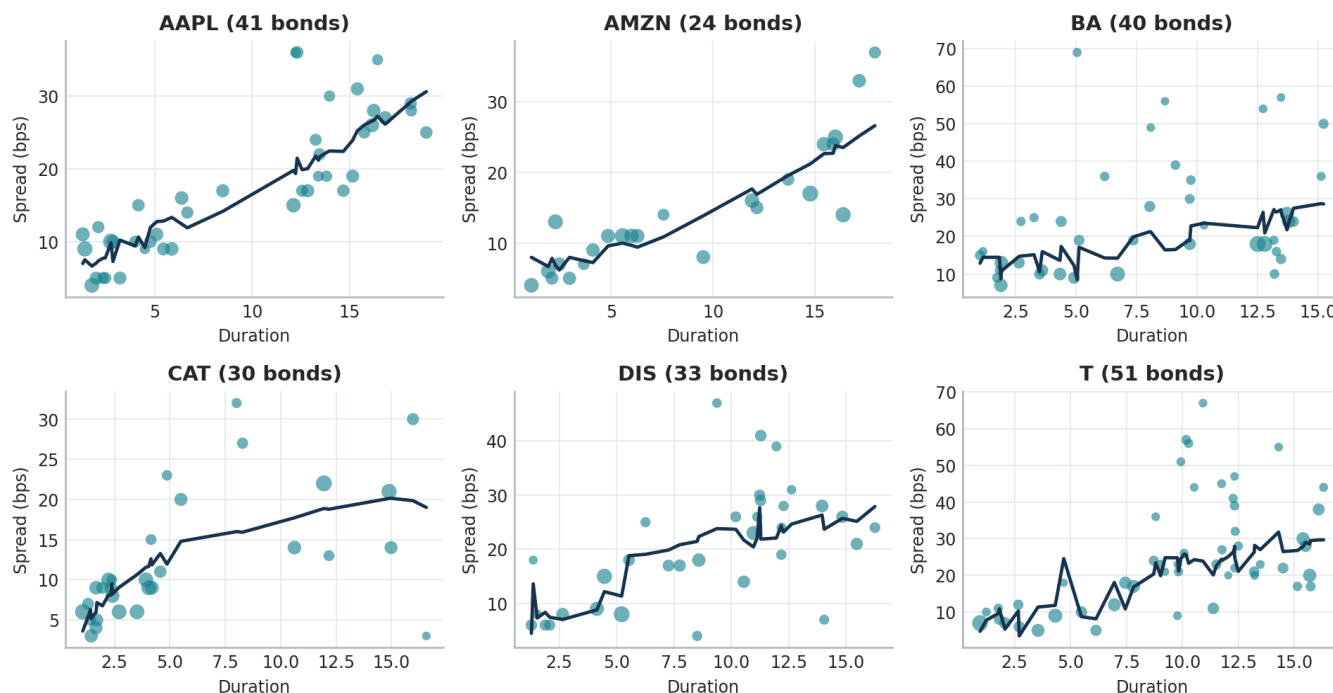
Issuer	Fit months	Average bonds	Median weighted R2	Median RMSE
AAPL	32	46.1	0.746	6.2 bps
AMZN	32	28.4	0.755	5.3 bps
BA	32	40.8	0.376	10.5 bps
CAT	32	29.9	0.617	7.1 bps
DIS	32	38.7	0.535	8.4 bps
T	32	52.4	0.386	12.8 bps

The signal is the residual: observed Treasury spread minus fitted spread. Positive residuals are classified as cheap and negative residuals as rich. Ranking is performed within issuer, which avoids comparing raw spread levels across firms with different credit risk.

3. Issuer Spread Curves

The plots below show the latest common cross-section, March 31, 2025. Marker size reflects TRACE dollar volume, and the fitted line is generated from the weighted regression.

Issuer spread curves on March 31, 2025



Reading the cross-sections

AAPL and AMZN have the strongest median fit statistics, while BA and T display more unexplained spread dispersion. The model does not assume that every residual is a mispricing. A large residual can also reflect liquidity, issue size, maturity, optionality, or missing security-level information.

This distinction matters for portfolio construction: the residual is a ranking signal, not a guarantee of convergence. The later outlier review checks whether selected bonds have systematically different liquidity and maturity characteristics.

4. Portfolio Construction

Each issuer is treated as a separate relative-value sleeve. The portfolio buys bonds with the widest positive residuals, shorts bonds with the most negative residuals, and scales the short side until the signed market-value duration contribution is approximately zero. Each sleeve is normalized to one dollar of gross market value and available sleeves are equal-weighted.

DURATION HEDGE

$$\text{sum}(\text{Position} \times \text{Percentage Price} \times \text{Duration}) = 0$$

Three specifications

Pass	Selection rule	Research question
1. Baseline top 3	Three cheapest and three richest; size by residual strength	Does the direct rich-cheap rule produce a payoff?
2. Concentrated top 2	Two strongest bonds on each side	Do weaker selected residuals dilute the signal?
3. Liquidity-aware top 3	Three per side; ranking incorporates TRACE dollar volume	Does implementation quality improve selection?

Rebalancing and costs

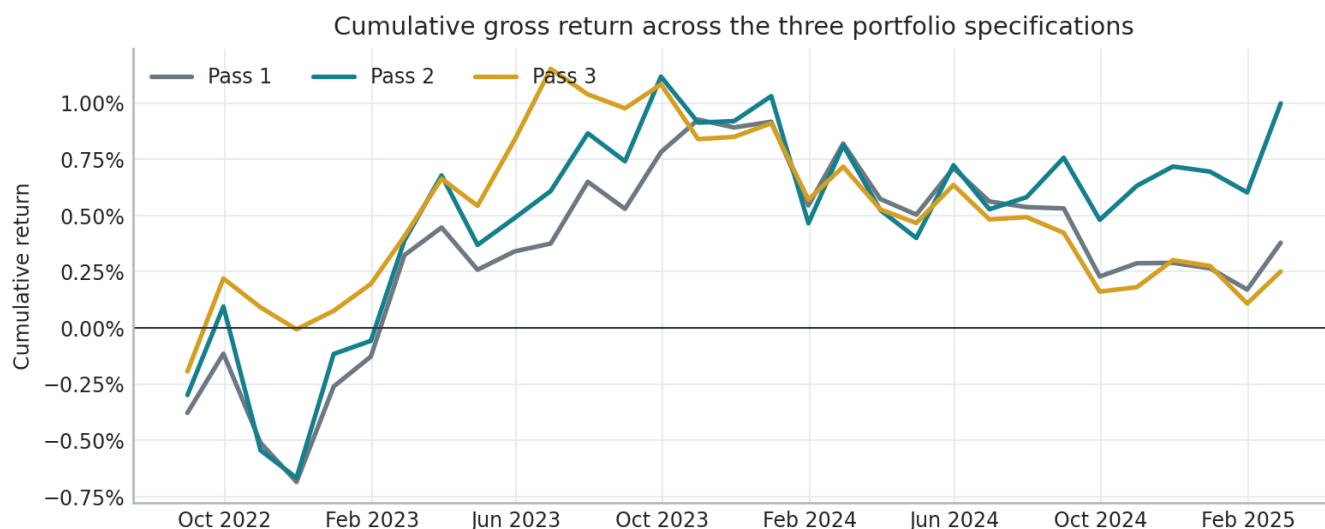
Portfolios rebalance monthly when a valid next-month return is available. Turnover is one half of the absolute change in bond market weights. Net returns subtract one-way trading costs of 0, 10, 25, or 50 basis points multiplied by turnover.

Risk that remains

DV01 neutrality removes first-order exposure to a parallel Treasury-rate move. It does not neutralize curve slope and curvature, spread duration, issuer beta, downgrade risk, liquidity risk, financing, or short-borrow constraints. These remaining exposures are important because the measured gross return is small.

5. Main Backtest Results

Pass 2 is the strongest specification in the main sample. It earns 0.39% gross annualized return with 1.02% volatility, a 0.38 gross Sharpe ratio, and 1.00% cumulative return over 31 months.



Portfolio	Gross ann.	Gross vol.	Sharpe	Total return	Win rate	Turnover	Net ann. 25 bp
1. Baseline top 3	0.15%	0.78%	0.19	0.38%	51.6%	0.78	-2.10%
2. Concentrated top 2	0.39%	1.02%	0.38	1.00%	58.1%	0.82	-1.98%
3. Liquidity-aware top 3	0.10%	0.64%	0.15	0.25%	51.6%	0.79	-2.17%

Issuer breadth

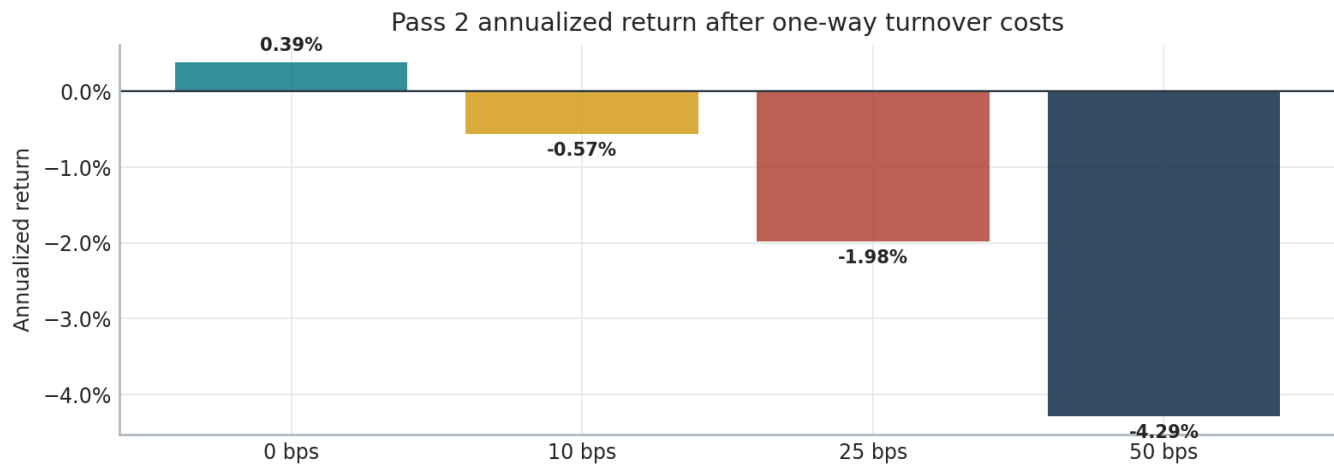
Issuer	Ann. return	Ann. vol.	Sharpe	Total return	Win rate
AAPL	0.39%	1.11%	0.35	1.01%	51.6%
AMZN	-0.39%	0.79%	-0.50	-1.02%	51.6%
BA	0.73%	1.90%	0.39	1.90%	48.4%
CAT	-0.60%	3.66%	-0.16	-1.55%	58.1%
DIS	1.66%	1.59%	1.05	4.36%	67.7%
T	0.42%	3.16%	0.13	1.08%	45.2%

DIS has the strongest issuer-level result, while AMZN and CAT are negative. The best combined month is December 2022 at 0.55%; the worst is October 2022 at -0.64%.

UNTREATED-DATA COMPARISON

Before excluding the isolated AAPL mark, gross annualized returns were 2.75%, 4.48%, and 5.99% for Passes 1, 2, and 3. These higher values are shown as a data-quality sensitivity on page 9 and are not used as the final estimates.

6. Costs and Outlier Review



Average one-way monthly turnover is 0.82. A 10 bp cost reduces annualized return to -0.57%; at 25 bp it falls to -1.98%. The portfolio is close to zero net DV01 by construction, but accurate hedging cannot compensate for a signal whose gross return is smaller than implementation cost.

Characteristics of selected bonds

Group	Obs.	Median abs. residual	Median dollar volume	Median amount out.	Median duration	Median TMT
Long Cheap	372	27.2 bps	\$17,578,816	\$500,000	12.68	20.23
Not Selected	6,814	4.2 bps	\$50,222,070	\$1,100,000	7.11	8.71
Short Rich	372	15.5 bps	\$21,082,954	\$758,680	13.66	24.25

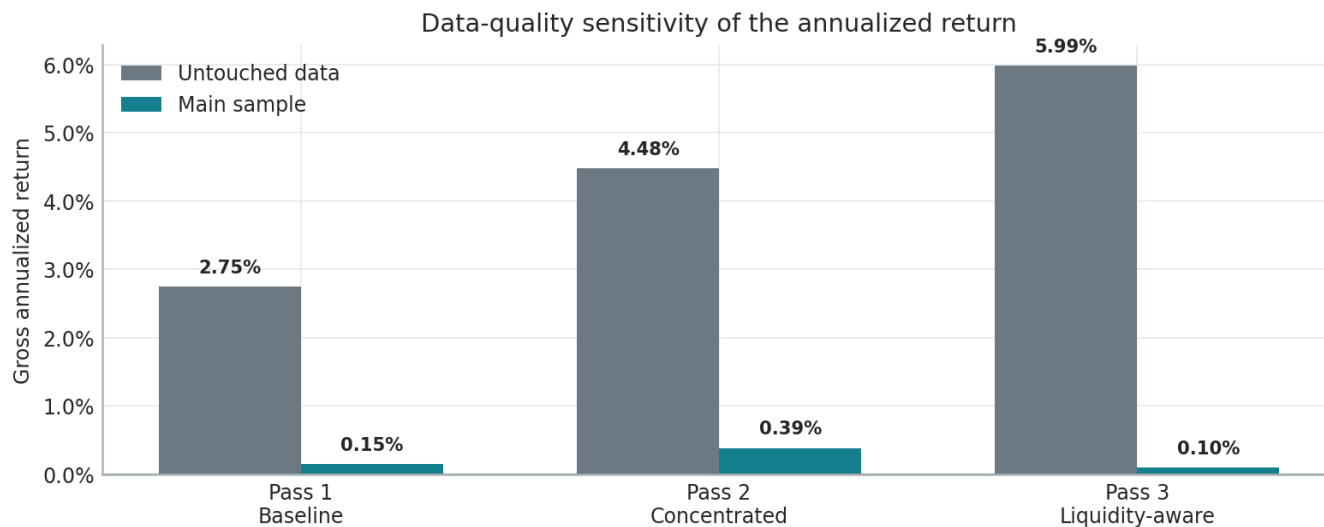
Long-cheap observations have lower median dollar volume, smaller issue size, and longer duration than non-selected observations. Part of the residual therefore appears to compensate investors for liquidity and long-dated risk rather than represent a pure pricing error.

IMPLEMENTATION CONCLUSION

The strongest gross specification is not economically attractive after plausible costs, and its selected bonds carry characteristics that may make actual execution more difficult than the backtest assumes.

7. Data-Quality Sensitivity

As a diagnostic, I rerun the same three portfolios on the otherwise identical data with the isolated-mark screen disabled. This comparison measures how strongly one month-end input can affect fitted residuals, bond selection, and reported returns.



Portfolio	Untouched gross	Main gross	Untouched net 25 bp	Main net 25 bp
1. Baseline top 3	2.75%	0.15%	0.45%	-2.10%
2. Concentrated top 2	4.48%	0.39%	2.02%	-1.98%
3. Liquidity-aware top 3	5.99%	0.10%	3.60%	-2.17%

Interpretation

With the February 2025 AAPL 037833BX7 mark left in place, gross annualized returns rise to 2.75%, 4.48%, and 5.99%. The same price enters the Apple curve and creates two extreme adjacent return intervals, so the higher numbers are a data-quality sensitivity rather than a second performance estimate.

USE IN THE FINAL ASSESSMENT

The main-sample results on page 7 are the performance findings. The untouched results are retained only to show why bond-level validation matters before a relative-value backtest is interpreted.

8. Conclusion

FINAL ASSESSMENT

The issuer-curve framework produces a coherent relative-value ranking and a well-controlled parallel-rate hedge, but the measured return does not survive plausible transaction costs. I would not present the strategy as profitable in this sample.

What the analysis supports

- The three-factor curve provides an interpretable way to compare bonds from the same issuer across the maturity spectrum.
- Concentrating on the strongest residuals performs better than the baseline and liquidity-aware alternatives in the main sample.
- DV01 scaling removes first-order parallel-rate exposure with negligible residual net DV01.
- Turnover costs and liquidity characteristics are large relative to the observed gross return.

Limitations and next steps

The data do not provide complete call schedules, bid-offer spreads, 144A status, subsidiary mapping, or borrow availability. The isolated-mark rule is an ex-post historical validation procedure. The three portfolio specifications are also compared on the same sample, which introduces selection bias.

A stronger next test would validate month-end prices against a second source, freeze the portfolio rule on an earlier training period, and evaluate later observations with point-in-time transaction costs, financing, borrow constraints, call information, and key-rate risk controls.

References

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